NETWORK WALKS

WEEK 2 – CYBERSECURITY TRAINING

FINAL PROJECT REPORT OF WEEK 2

Project Areas:

- PM1 – Web Footprinting and Information Gathering
- PM5 – Zenmap Network Scanning and Host Discovery

Submitted By: Harini Madasu
Training Organization: Network Walks

---

1. INTRODUCTION

Cybersecurity involves protecting digital systems, networks, applications, and information from unauthorized access and potential security threats. Understanding how information is gathered about websites and how devices communicate within networks is an important part of developing cybersecurity skills.

As part of Week 2 of the Network Walks cybersecurity training, I completed two practical projects: Web Footprinting and Information Gathering (PM1), and Zenmap Network Scanning and Host Discovery (PM5).

The first project focused on understanding domain information gathering, DNS analysis, website technology identification, and Web Application Firewall detection. The second project introduced network discovery, port scanning, service detection, and operating system detection using Zenmap and Nmap.

Both projects were performed in a Kali Linux environment using commonly used cybersecurity tools. The exercises provided practical exposure to reconnaissance techniques and basic network security assessment.

2. OBJECTIVES

The primary objectives of Week 2 were:

1. To understand the fundamentals of web footprinting and network reconnaissance.
2. To gain practical experience with commonly used cybersecurity tools.
3. To understand domain registration information and DNS infrastructure.
4. To explore website technology identification and WAF detection.
5. To learn the fundamentals of network scanning and host discovery.

6. To understand TCP port scanning and service version detection.
7. To explore operating system detection and scan result analysis.
8. To develop the ability to interpret technical outputs and document practical observations.

3. ENVIRONMENT AND TECHNOLOGIES USED

The practical exercises were conducted using Kali Linux installed in an Oracle VirtualBox virtual machine.

Component	Details
Operating System	Kali Linux
Virtualization Platform	Oracle VirtualBox
Programming / Command Environment	Linux Terminal
Footprinting Tools	WHOIS, WhatWeb, Nslookup, cURL, Wafw00f, DNSRecon
Network Scanning Tools	Zenmap and Nmap
Demonstration Domain	example.com
Network Scanning Target	127.0.0.1 (Localhost)

4. PROJECT OVERVIEW

4.1 PM1 – Web Footprinting and Information Gathering

Objective: To understand how publicly available information about a website and its domain infrastructure can be collected using different footprinting tools.

This project involved exploring six tools: WHOIS, WhatWeb, Nslookup, cURL, Wafw00f, and DNSRecon.

The practical exercises covered domain registration lookup, website technology identification, DNS record queries, HTTP response header inspection, Web Application Firewall detection, and DNS enumeration.

Key Learning Areas:

- Domain registration information gathering.
- DNS records and domain name resolution.

- Website technology fingerprinting.
- HTTP response header analysis.
- Web Application Firewall identification.
- DNS enumeration techniques.

Project Documentation: Refer to the PM1-Footprinting folder in the GitHub repository for detailed commands, practical observations, and screenshots.

4.2 PM5 – Zenmap Network Scanning and Host Discovery

Objective: To understand network scanning techniques and examine host availability, TCP ports, services, and operating system detection.

This project involved using Zenmap, the graphical interface of Nmap, to perform network scanning exercises against the local loopback address (127.0.0.1).

The activities included exploring Zenmap's scanning interface, performing Ping and Regular Scans, examining scanning profiles, executing custom service detection commands, attempting operating system detection, and saving scan results.

Key Learning Areas:

- Host discovery and availability checking.
- TCP port scanning.
- Understanding open and closed port states.
- Service version detection.
- Operating system fingerprinting.
- Saving and interpreting scan results.

Project Documentation: Refer to the PM5-Zenmap-Network-Scanning folder in the GitHub repository for detailed commands, practical observations, and screenshots.

5. KEY RESULTS AND OBSERVATIONS

The two projects provided practical exposure to different aspects of cybersecurity reconnaissance and network assessment.

5.1 Footprinting Results

- WHOIS demonstrated how domain registration information can be retrieved.
- WhatWeb introduced website technology fingerprinting.
- Nslookup demonstrated how DNS records can be queried.

- cURL introduced HTTP response header inspection.
- Wafw00f identified Cloudflare protection for the demonstration domain.
- DNSRecon provided practical exposure to standard DNS enumeration.

These exercises demonstrated how different tools can collect different types of information about a domain and its associated infrastructure.

5.2 Network Scanning Results

- The Ping Scan successfully identified the localhost as an active host.
- The Regular Scan examined 1,000 TCP ports and reported all of them as closed.
- Service version detection was performed using Nmap's -sV option.
- Operating system detection was attempted using the -O option, but no definitive OS identification was obtained.
- The scan results were successfully saved in a text file for documentation and future reference.

These observations helped explain how network scanning results are interpreted and why the availability of open ports and network responses can affect further detection.

6. SKILLS AND KNOWLEDGE ACQUIRED

Through the Week 2 practical exercises, I developed a foundational understanding of the following concepts:

1. Information Gathering: Learned how domain and website information can be collected using reconnaissance tools.

2. DNS Analysis: Understood the purpose of DNS records and how DNS lookup utilities retrieve domain information.

3. Web Technology Identification: Gained exposure to identifying technologies and frameworks associated with websites.

4. Security Mechanism Awareness: Understood the role of Web Application Firewalls in protecting web applications.

5. Network Discovery: Learned how to check host availability and explore network scanning techniques.

6. Port and Service Analysis: Understood the fundamentals of TCP port states and service version detection.

7. Operating System Detection: Explored how Nmap attempts to identify operating systems through network responses.

8. Technical Documentation: Improved the ability to execute commands, interpret outputs, capture screenshots, and document practical work using GitHub.

7. CONCLUSION

The Network Walks Week 2 cybersecurity training provided valuable hands-on experience in web footprinting, information gathering, and network scanning.

Through PM1, I explored different tools used for collecting domain information, examining DNS records, identifying website technologies, and understanding web security mechanisms.

Through PM5, I gained practical exposure to Zenmap and Nmap for host discovery, TCP port scanning, service version detection, and operating system detection.

The exercises strengthened my understanding of cybersecurity reconnaissance and basic network assessment methodologies. They also helped improve my familiarity with the Kali Linux environment and commonly used cybersecurity utilities.

Overall, Week 2 was a valuable learning experience that enhanced my practical knowledge and strengthened my foundation in cybersecurity and network security.

8. REFERENCES

1. Kali Linux Official Website – https://www.kali.org/
2. Nmap Official Documentation – https://nmap.org/
3. Zenmap Documentation – https://nmap.org/zenmap/
4. ICANN WHOIS Information – https://www.icann.org/resources/pages/whois-2018-01-17-en
5. WhatWeb GitHub Repository – https://github.com/urbanadventurer/WhatWeb
6. cURL Official Website – https://curl.se/
7. Wafw00f GitHub Repository – https://github.com/EnableSecurity/wafw00f
8. DNSRecon GitHub Repository – https://github.com/darkoperator/dnsrecon